

# Physics Publications with Major Personal Contributions

Oliver Gutsche

May 08, 2026

1. A portrait of the Higgs boson by the CMS experiment ten years after the discovery.

Tumasyan, Armen et al.

[Nature 607 \(2022\) 60](#) | [arXiv:2207.00043](#) | [DOI:10.1038/s41586-022-04892-x](#) | 2022-06-30

2. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton-proton collisions at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 280](#) | [arXiv:1811.06562](#) | [DOI:10.1140/epjc/s10052-019-6730-7](#) | 2018-11-15

3. Search for dark matter in events with energetic, hadronically decaying top quarks and missing transverse momentum at  $\sqrt{s}=13$  TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 027](#) | [arXiv:1801.08427](#) | [DOI:10.1007/JHEP06\(2018\)027](#) | 2018-01-25

4. Measurements of  $t\bar{t}$  charge asymmetry using dilepton final states in pp collisions at  $\sqrt{s}=8$  TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 365](#) | [arXiv:1603.06221](#) | [DOI:10.1016/j.physletb.2016.07.006](#) | 2016-03-20

5. Measurements of  $t\bar{t}$  spin correlations and top quark polarization using dilepton final states in pp collisions at  $\sqrt{s} = 8$  TeV

Khachatryan, V. et al.

[Phys.Rev.D 93 \(2016\) 052007](#) | [arXiv:1601.01107](#) | [DOI:10.1103/PhysRevD.93.052007](#) | 2016-01-06

6. Measurements of the  $t\bar{t}$  Charge Asymmetry Using the Dilepton Decay Channel in pp Collisions at  $\sqrt{s} = 7$  TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2014\) 191](#) | [arXiv:1402.3803](#) | [DOI:10.1007/JHEP04\(2014\)191](#) | 2014-02-16

7. Measurements of  $t\bar{t}$  Spin Correlations and Top-Quark Polarization Using Dilepton Final States in pp Collisions at  $\sqrt{s} = 7$  TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 182001](#) | [arXiv:1311.3924](#) | [DOI:10.1103/PhysRevLett.112.182001](#) | 2013-11-15

8. Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 30](#) | [arXiv:1207.7235](#) | [DOI:10.1016/j.physletb.2012.08.021](#) | 2012-07